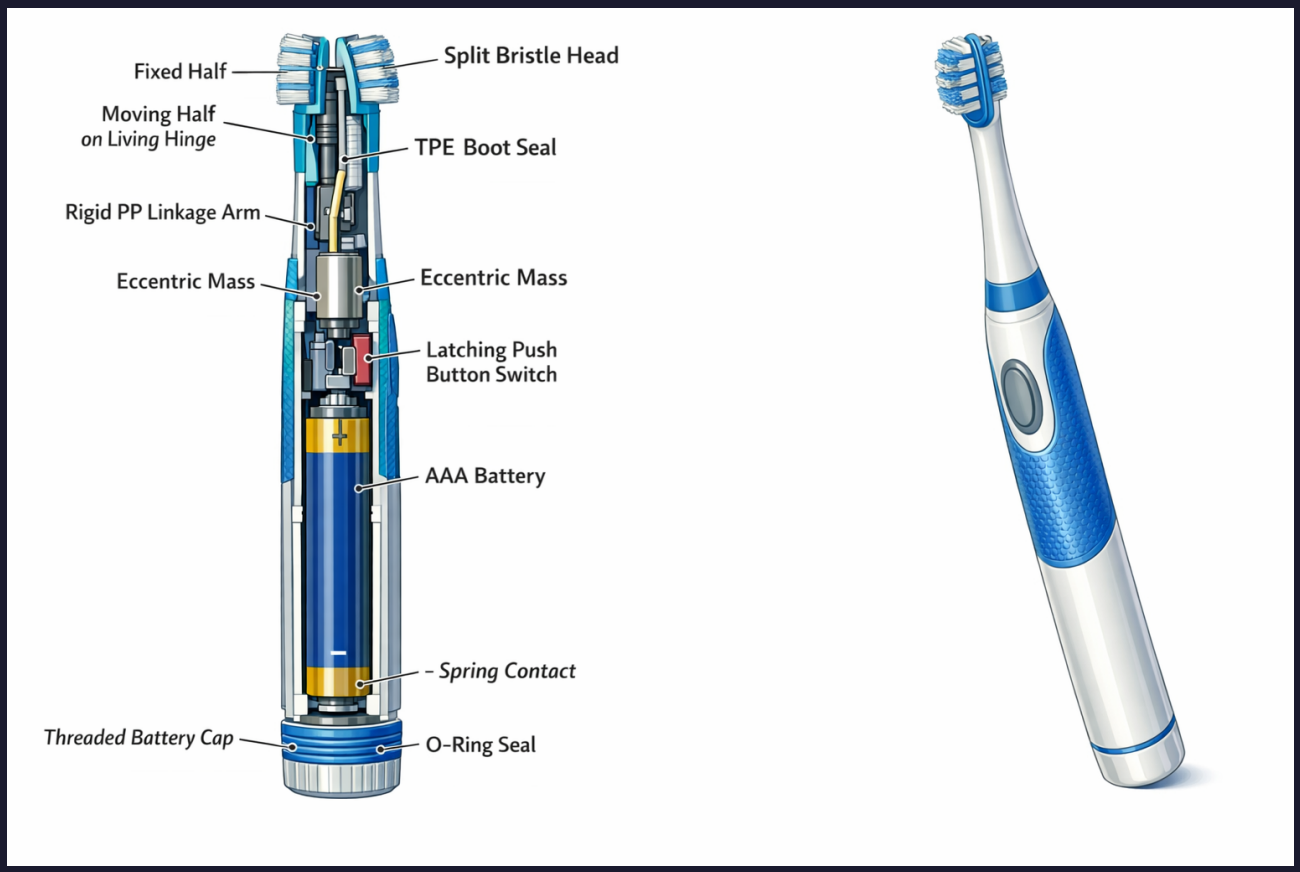
# Consumable Electric Toothbrush

## Use it. Toss it. Open a new one.

A battery-powered vibrating toothbrush designed as a true consumable. AAA alkaline, no MCU, dual-injection sealed. $3-5 retail vs. Oral-B Pulsar at $8-12.



# The Problem

## Oral-B Pulsar is overpriced for a disposable

At $8-12, the Pulsar pretends to be reusable but has a sealed, non-replaceable battery. When it dies, you throw it away anyway. You're paying premium prices for a disposable product.

## Manual brushes don't clean as well

Powered vibration removes significantly more plaque than manual brushing. But rechargeable electric brushes cost $30-200 plus replacement heads. Budget consumers are stuck with manual.

## No honest sub-$5 powered option exists

The market has premium rechargeable brushes ($30+) and the Oral-B Pulsar ($8-12). Nothing fills the gap below $5 -- a price point where powered brushing becomes truly disposable.

### TARGET USERS

**Budget-conscious consumers | Travelers**

**Hotels & hospitality | Multi-pack buyers**

# How It Works

## 1 Open

**Tear open blister pack**

AAA alkaline battery is pre-installed with years of shelf life. No charging, no pairing, no setup of any kind.

## 2 Press

**Push the button once**

Latching push-button switch turns motor on. ERM vibration motor spins up instantly. Bristles oscillate via the split-head mechanism.

## 3 Brush

**2 minutes, twice daily**

Split bristle head concentrates oscillation at the bristle tips for effective plaque removal. Rinse under tap after use. IPX5 sealed.

## 4 Toss

**~90 days later, motor stops**

Alkaline cell drops below motor stall voltage. Clear end-of-life: it works or it doesn't. Discard and open a new one.

# Architecture

## Power Path

**AAA Cell --> Latching Switch --> ERM Motor**

1.5V alkaline  push on/off  60-100mA, direct drive

## Mechanical Path

**Motor --> Eccentric --> Linkage --> Split Head --> Bristles**

## SUBSYSTEMS

### AAA Alkaline Cell (1.5V)

1000-1200 mAh, ~750 min runtime, years of shelf life

### Latching Push-Button

Mechanical push-on/push-off, sealed by TPE overmold

### Cylindrical ERM Motor

6x12mm, 60-100mA at 1.5V, axial mount in handle neck

### Split Bristle Head

Fixed + moving halves, living hinge pivot, 1-2mm oscillation

### Rigid PP Body (1st shot)

Handle, motor pocket, battery tube, insert-molded bristles

### TPE Overmold (2nd shot)

Seals all penetrations, grip texture, button membrane, head boot

### Battery Cap + O-ring

Threaded PP cap, static radial seal, factory-installed

## Zero software. Zero electronics.

No MCU, no PCB, no firmware, no app, no cloud. The entire electrical system is 3 components and 2 wires.

# Dual-Injection Sealing

How do you waterproof a $1.50 product for 90 days of wet use?

## 1st Shot: Rigid PP Body

Injection-mold the structural handle with bristles pre-loaded in the mold cavity. Plastic flows around each bristle base, creating a watertight seal at every tuft. No secondary sealing.

## 2nd Shot: TPE Overmold

After motor, wiring, switch, and battery are assembled into the rigid body, the 2nd injection encapsulates everything. TPE bonds chemically to PP, sealing all penetrations in one step: grip texture, button membrane, and dynamic head boot.

## 4 Seal Zones

1. **Bristle insert-mold**
   PP flows around nylon tuft bases during 1st injection
2. **TPE body overmold**
   2nd shot seals handle, switch, wire penetrations
3. **TPE head boot**
   Flexible boot around split-head pivot, survives millions of flex cycles
4. **O-ring battery cap**
   Threaded PP cap compresses O-ring at handle base

# Constraints & BOM

**BOM cost** — **< $1.50**

At 10k units; < $1.20 at 50k

**Motor runtime** — **>= 350 min**

~750 min calculated (AAA @ 80mA)

**Water resistance** — **IPX5**

Dual-injection + O-ring, 90 days wet use

**Handle diameter** — **<= 16mm**

AAA cell (10.5mm) + PP wall + TPE

**Retail price** — **$3 - $5**

Undercutting Oral-B Pulsar ($8-12)

**Shelf life** — **> 2 years**

Alkaline chemistry, sealed blister pack

## BOM ESTIMATE (10k units)

| | |
|---|---|
| ERM motor (cylindrical, 6x12mm) | **$0.12** |
| AAA alkaline cell | **$0.07** |
| Latching push-button switch | **$0.03** |
| Wiring (2x, 26AWG, tinned) | **$0.01** |
| Spring contact + plate contact | **$0.02** |
| PA-612 nylon bristle tufts (x35) | **$0.02** |
| O-ring (battery cap seal) | **$0.01** |
| PP resin (1st shot, ~8g) | **$0.02** |
| TPE resin (2nd shot, ~3g) | **$0.02** |
| Battery cap (threaded PP) | **$0.02** |
| Blister packaging + card | **$0.08** |
| Assembly + test (90-100s/unit) | **$0.72** |
| **Total COGS** | **~$1.14 (50k: ~$0.92)** |

# Hardest Problems

## 1 Dynamic seal at the split head joint

The moving bristle half oscillates ~1-2mm at 150+ Hz while the TPE boot must keep water out of the motor cavity. This is a fatigue + sealing problem -- the boot must survive millions of flex cycles in a wet, toothpaste-laden environment without cracking or delaminating from the PP substrate.

## 2 Insert-molding bristle seal quality at speed

Each brush head has ~35 bristle tufts penetrating the PP surface. At production speed (seconds per unit), every tuft base must be fully sealed by the injection process. A single unsealed tuft is a water ingress path. Narrow process window for injection pressure, temperature, and bristle positioning.

## 3 BOM discipline with dual-injection process

Overmolding adds ~$0.10-0.20 per unit vs. single-shot molding, and tooling is 2-3x higher ($15k-25k). With $3-5 retail and distribution margins to cover, every component must be ruthlessly cost-optimized. No room for any extras.

All three problems are manufacturing challenges, not design unknowns. The Oral-B Pulsar proves the architecture works -- the risk is execution at lower cost.

# Gate Result & Next

**GATE: PASS**

40 pass / 47 N/A / 2 fail

System description complete.

**Ready for mold design.**

## POWER

80 mA active | < 1 uA off | AAA 1000 mAh

~750 min runtime | ~187 days @ 4 min/day | non-rechargeable

## KEY SPECS

~160mm long | ~15mm dia | IPX5 | Single SKU

No MCU | No PCB | No app | $3-5 retail

## 7 OPEN ITEMS

| | |
|---|---|
| M1 | Motor performance validation at end-of-life voltage (~1.0V) |
| M2 | Living hinge geometry: thickness, width, PP grade for 4M cycles |
| M2 | TPE head boot fatigue life in wet toothpaste environment |
| M3 | Insert-mold bristle seal QC: tuft pull-force per ISO 20126 |
| M3 | Retail channel acceptance test (shelf appeal, blister design) |
| M4 | O-ring compression verification for production QC |
| M4 | Motor current draw screening for dead/high-draw units |

Next step: source motor samples and AAA cells, build a hand-assembled prototype in a 3D-printed shell, and validate the split-head oscillation and motor stall voltage. Then commit to dual-injection tooling.